



28-STYLE-LEARNING-SPEC.md

— Editing like this photographer, not like an AI



Generic AI editing is a commodity and it is already free. The entire defensible value of AURA is that after three weddings it edits like **you**. This file specifies how that happens, how it is measured, and how it fails safely.

1. What style actually is

Photographers describe style in words like "warm and filmic" or "moody". Those words are unmeasurable. Decompose style into five learnable components, each with its own representation, evaluation and failure mode.

Component	What is learned	Signal source	Data needed
Selection	Which frames are keepers	Their historical cull decisions	2 weddings
Tone	Exposure, contrast curve, black point, highlight recovery	Paired RAW plus their edited JPEG	300 pairs
Colour	White balance bias, saturation by hue, skin rendering, split toning	Same pairs, in Lab	300 pairs
Crop	Aspect preference, headroom, horizon discipline	Crop deltas between RAW and export	200 crops
Narrative	Which moments matter, how many of each, gallery pacing	Delivered gallery sequence and counts	3 galleries

Keep them separate. A photographer may love the tone matching and hate the crop suggestions, and a single monolithic style model makes that impossible to express. Separate components also mean a failure in one degrades one thing, not everything.

2. The style profile, frozen contract

```
// crates/aura-style/src/contract/profile.rs -- FROZEN CONTRACT
pub struct StyleProfile {
    pub profile_uuid: Uuid,
    pub photographer_id: Uuid,
    pub version: u32,
    pub selection: SelectionStyle,
    pub tone: ToneStyle,
    pub colour: ColourStyle,
    pub crop: CropStyle,
    pub narrative: NarrativeStyle,
    pub evidence: Evidence,
    pub created_at: DateTime<Utc>,
    pub parent_version: Option<u32>,
}

pub struct Evidence {
    pub weddings_learned_from: u32,
    pub image_pairs: u32,
    pub overrides_observed: u32,
    pub per_component_confidence: BTreeMap<String, f32>,
    pub last_updated: DateTime<Utc>,
}

pub struct ToneStyle {
    pub exposure_bias_ev: f32,
    pub contrast_curve: [f32; 9],
    pub black_point: f32,
```

```
pub highlight_rolloff: f32,  
pub confidence: f32,  
pub sample_count: u32,  
}
```

Every component carries its own confidence and sample count. A component below its minimum sample threshold is simply not applied, and the UI says so plainly rather than guessing.

3. How a profile is built

Stage 1, cold start. No profile. Use the shipped baseline and the optional 30-image calibration from first run. Confidence is low everywhere; autonomy stays in suggest mode.

Stage 2, learning from history. Ask for a folder of past weddings containing both RAWs and delivered JPEGs. This is the highest-value onboarding action in the product and it should be asked for at the right moment, which is after the first successful run, not before.

Parameter recovery from a RAW and edited JPEG pair is the technical heart of this file:

1. Align the pair, accounting for crop and rotation.
2. Convert both to a linear working space.
3. Fit tone by regressing the luminance transfer curve on a sampled grid.
4. Fit colour by solving for the white balance and per hue saturation and shift that minimise deltaE in Lab, with faces masked and weighted separately.
5. Record residual error. A high residual means the edit contained local adjustments AURA cannot represent, and that pair must be down weighted rather than fitted badly.

Stage 3, learning from corrections. Every override is a labelled example. Weight recent overrides higher, and require a pattern before changing anything: three consistent corrections in the same direction, not one.

Stage 4, steady state. The profile updates after each completed wedding, versioned, never overwritten.

4. Rules that keep this trustworthy

- **Never learn silently.** Any profile change is announced: what changed, based on what evidence, with a one click revert.
- **Never learn from an unfinished project.** A half culled wedding is not a judgement, it is an interruption.
- **Never learn across photographers.** Studio accounts keep separate profiles per shooter, because the second shooter's style is not the lead's.
- **Never let learning break determinism.** The profile version is part of the render fingerprint, so the same photo plus the same profile version always produces the same pixels. A profile update creates a new version; it never silently changes yesterday's output.
- **Never learn a bias into skin tone.** Colour learning is constrained so that no learned transform can shift skin lightness beyond a bounded range. Style is not a licence to change what people look like.
- **Always allow a hard reset.** One button, back to the shipped baseline, no questions asked.

5. Evaluation

Style quality is measured, not asserted.

Metric	Definition	Ship gate
Selection agreement	F1 against the photographer's own cull on a held out wedding	Above their own week to week self consistency
Tone match	Median deltaE2000 against their edit, held out pairs	Under 3.0, and under 2.0 on faces
Override rate	Fraction of AURA edits the photographer changes	Falls across the first five weddings; a flat line is a failure
Blind preference	Photographer picks between their edit and AURA's, unlabelled	At least 40 percent choose AURA by wedding three
Consistency	Variance of applied tone within one gallery	No higher than their own galleries

The self consistency baseline matters more than any absolute number. A photographer does not edit identically two weeks apart; measure that first, and never demand that the model beat a standard the human does not meet.

6. Failure modes

Failure	Cause	Guard
Overfits one wedding	A single golden hour beach wedding teaches warm tone everywhere	Require variety across venue and lighting before raising confidence; per condition profiles
Learns from bad pairs	JPEGs were re edited by a client or compressed by a gallery host	Residual threshold; reject pairs above it and report how many were rejected
Drifts	Small corrections compound over a season into something the photographer never chose	Version history, drift alert when cumulative change passes a threshold, one click revert to any version
Style collapse	Everything converges to a safe average	Variance floor: the applied variation must not fall below their historical variation
Cold start pretends	Confident edits with no evidence	Components below sample threshold are not applied at all

7. Tests

- ☐ Round trip: synthesise an edit with known parameters, recover them, assert within tolerance.
- ☐ Held out wedding: train on two, evaluate on a third, report all five metrics.
- ☐ Determinism: the same photo with the same profile version renders identical bytes across ten runs.
- ☐ Skin safety: across the full Monk scale, no learned profile shifts skin lightness beyond the bound.
- ☐ Bad pair rejection: feed deliberately re compressed and re edited JPEGs, assert they are rejected and counted.

- ☐ Revert: apply five profile versions, revert to version two, assert renders match version two exactly.
- ☐ Cold start honesty: with no evidence, assert no tone or colour transform is applied and the UI states why.
- ☐ Cross contamination: two profiles on one machine never influence each other.

8. Acceptance criteria

- ☐ `profile.rs` matches section 2 and is under contract freeze.
- ☐ Profile version is part of the render fingerprint.
- ☐ Every component reports confidence and sample count in the UI, in plain language.
- ☐ Every profile change is announced and revertible.
- ☐ Self consistency baseline measured for at least ten photographers before any ship gate is set.
- ☐ Override rate demonstrably falls across five consecutive weddings for at least five beta users.
- ☐ Hard reset returns byte identical output to the shipped baseline.

9. Claude Code prompt

Act as `MLL` (ML Lead) with `COL` (Colour Science) and `SRG` (Rendering) reviewing. Implement `28-STYLE-LEARNING-SPEC.md`: the frozen `StyleProfile` contract, the paired RAW and JPEG parameter recovery in section 3 including alignment and residual rejection, per component confidence gating, versioned profiles with revert, the drift detector, and the skin safety constraint.

Constraints: no component may be applied below its sample threshold; the profile version must enter the render fingerprint; no learned transform may alter skin lightness beyond the bound; learning must never run on an unfinished project.

Then run the round trip and held out wedding tests, paste all five metrics, and show the determinism test passing across ten runs with the same profile version. Report how many training pairs were rejected by the residual threshold and why.